A-A-53909
26 June 1989

COMMERCIAL ITEM DESCRIPTION

DRILL, PULP CANAL, DENTAL (Gates Glidden)

The General Services Administration has authorized the use of this Commercial Item Description as a replacement for Military Specification MIL-D-37775B, which is cancelled.

Salient characteristics:

This commercial item description covers Gates Glidden drills suitable for use in endodontic procedures. The drills covered by this item description shall be of the following sizes as specified herein:

Size - 1
Size - 2
Size - 3
Size - 4
Size - 5
Size - 6

Material. The drills shall be fabricated from a chromium-molybdenum-manganese stainless steel, having the following composition:

Element	Percent
Carbon	0.20 - 0.25
Silicon	0.50 - 0.70
Manganese	1.50 - 1.85
Phosphorus	Max 0.030
Sulfur	0.018 - 0.25
Chromium	12.00 - 14.00
Nickel	0.75 - 1.15
Molybdenum	1.10 - 1.30

Style, design and dimensions. The drills shall be suitable for the enlargement, shaping and debridement of the dental pulp chamber and canal when performing endodontic treatment. The drills shall be the Gates Glidden style, sizes 1 through 6, suitable for use in the latch contra-angle handpieces. The drills shall be of one-piece construction, consisting of a shank, an elongated neck, and a drill head (operative end). The Gates Glidden drills, shall conform to the Defense Personnel Support Center (DPSC) Drawing 23309, for sizes 1 through 6

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Hardness. The operative end and neck of the drills shall have a Vickers hardness (Diamond pyramid hardness) of 500 to 680. The hardness shall be determined in accordance with the requirements as specified herein.

Tensile strength. The tensile strength of the raw drill wire for the drills shall be 80 to 110 kilograms per square millimeter. Tensile strength shall be tested as specified herein.

Sterilization. The drills shall show no evidence of corrosion after having been subjected to the sterilization method specified herein.

Runout. The total indicated runout of the neck of the drills shall not exceed 0.003 inch, when tested as specified herein.

Marking. The shank of the drills shall be permanently and legibly marked to identify the size of the drill, such as a series of circular grooves corresponding to the size of the drill. Marking with manufacturer's name or trademark is optional.

Finish. The blade edge of the operative end shall have a sharp honed finish. The neck and latch end shall have a black finish, except for the shank portion, which shall have a bright finely machined finish.

Workmanship. The drills shall be free from defects which detract from their appearance or impair their serviceability.

Unit. Package (PG). One package containing six drills of the style specified, constitutes one unit.

Quality Assurance Provisions.

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

Certificates of quality. Certificates of quality, supplied by the manufacturer of the steel may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government.

Inspection. Inspection, as used in this item description, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

Examination. The drills shall be examined for defects including, but not limited to those listed below:

Defects

Size of drill not as specified.
Design and dimensions of drill not as specified.
Finish (fine machined and properly sharpened) of operative end of drill not as specified.
Latch portion of drill not cleanly machined or not free of burrs.
Shank of drill not smoothly finished.

The examination shall be conducted at inspection level II, except for dimensions which shall be conducted at inspection level S-1, AQL 2.5 (percent defective). The unit of product for examination purposes shall be one drill.

Testing. Sampling for tests for hardness and sterilization shall be conducted at inspection level S-1, AQL 1.0 (percent defective). Runout shall be conducted at inspection level S-4, for chemical analysis, a sufficient quantity of each lot of steel used in the fabrication of the drills shall be selected, and for tensile strength a minimum of three specimens from each lot of raw drill wire shall be tested. Acceptance number shall be zero (0) for all sample sizes for runout, chemical analysis and tensile strength.

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Tests. Tests shall be conducted to determine compliance with item description requirements.

Chemical analysis. In the absence of certificates of quality for the steel, chemical analysis shall be conducted in accordance with Method 111 or 112 of FED-STD-151.

Hardness. Hardness testing shall be performed in accordance with ASTM E92 for Vickers hardness.

Tensile strength. Tensile strength testing shall be performed in accordance with ASTM A370.

Sterilization. The test drills shall be cleaned in accordance with the instructions supplied by the manufacturer. The drills shall then be steam autoclaved at 121°C, 15 psi for 30 minutes for 3 cycles. After the drills are removed from the sterilizer, the entire surface area of the drill shall be examined for evidence of corrosion, and tested for hardness.

Runout. The drill shall be placed and held in a "V" block or precision chuck and rotated slowly. A dial indicator sensitive to 0.0002 inch and equipped with a knife-edge foot, a shadow-graph, or other suitable equipment of equal accuracy, shall be employed. If a dial indicator is used, the indicator shall be mounted with the shaft perpendicular to the axis of the drill, and so positioned that the foot contacts the drill neck at the point nearest the head. The maximum and minimum readings obtained during the complete 360-degree rotation of the drill shall be measured and recorded to the nearest 0.0005 inch. The difference between these two readings shall be taken as the total indicated runout.

Contractor certification. The contractor shall certify that the product offered meets the salient characteristics of this description and conforms to the producer's own drawings, specifications, standards, and quality assurance practices. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

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Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch/pounds units, providing they fall within the tolerances specified using conversion tables contained in the latest revision of Federal Standard No. 376, and all other requirements of this document are met. If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch/pound units, a request should be made to the contracting officer to determine if the product is acceptable. The contracting officer has the option of accepting or rejecting the product.

Regulatory requirements.

Federal Food, Drug and Cosmetic Act. If the product covered by this document has been determined by the U.S. Food and Drug Administration to be under its jurisdiction, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal Food, Drug and Cosmetic Act, as amended, and regulations promulgated thereunder. In addition, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of all other applicable Federal, State, and local statutes, ordinances, and regulations.

Recovered materials. The offeror/contractor is encouraged to use recovered material in accordance with Public Law 94-580, as amended, and as implemented by Federal Acquisition Regulation Subpart 23.4 to the maximum extent practical.

Preservation, packaging, packing, labeling, and marking. Unless otherwise specified, preservation, packaging and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling, etc., involving shipment from the supply source to the receiving activity, plus reshipment from receiving activity, and shall conform to applicable carrier's rules and regulations. Intermediate and exterior package quantities and labeling and marking shall be as specified in the contract and/or order.

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NOTE: The following National Stock Numbers are covered by this document:

National Stock Numbers	Item Identification
6520-01-011-2916	DRILL, PULP CANAL, DENTAL, Gates-Glidden, Angle Handpiece, Tungsten Steel, Size 1, 6s
6520-01-011-2917	DRILL, PULP CANAL, DENTAL, Gates-Glidden, Angle Handpiece, Tungsten Steel, Size 2, 6s
6520-01-011-2918	DRILL, PULP CANAL, DENTAL, Gates-Glidden, Angle Handpiece, Tungsten Steel, Size 3, 6s
6520-01-011-2919	DRILL, PULP CANAL, DENTAL, Gates-Glidden, Angle Handpiece, Tungsten Steel, Size 4, 6s
6520-01-011-2920	DRILL, PULP CANAL, DENTAL, Gates-Glidden, Angle Handpiece, Tungsten Steel, Size 5, 6s
6520-01-011-2921	DRILL, PULP CANAL, DENTAL, Gates-Glidden, Angle Handpiece, Tungsten Steel, Size 6, 6s

Ordering data. Intermediate/exterior package quantities, labeling and marking must be as specified in the contract and/or order.

MILITARY INTERESTS:

Custodians:

Army-MD
Navy-MS
Air Force-03

PREPARING ACTIVITY:

DoD-MB

Agent:

DLA-DM

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CIVIL AGENCY COORDINATING ACTIVITIES:

VA-OSS
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FDA-MPQAS

PROJECT NO. 6520-2237
LOCATION: NCR DISK 1

05780

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
A-A-53909	DRILL, PULP CANAL, DENTAL (Gates-Glidden)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*. _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording.

b. Recommended Wording.

c. Reason/Rationale for Recommendation

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

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NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

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